



Land & Water

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Crop Information

Tobacco

Maize

Sorghum

Wheat

Cotton

Alfalfa

Banana

Citrus

Grape

Pineapple

Potato

Groundnut

Olive

Safflower

Soybean

Sunflower

Bean

Pea

Sugarbeet

Sugarcane

Cabbage

Onion

Pepper

Tomato

Watermelon

Maize

This section presents information on water relations and water management of maize and provides links to other sources of information.

Crop Description and Climate

Maize (*Zea Mays*) originates in the Andean region of Central America. It is one of the most important cereals both for human and animal consumption and is grown for grain and forage. Present world production is about 594 million tons grain from about 139 million ha (FAOSTAT, 2000).

The crop is grown in climates ranging from temperate to tropic during the period when mean daily temperatures are above 15°C and frost-free. Adaptability of varieties in different climates varies widely. Successful cultivation markedly depends on the right choice of varieties so that the length of growing period of the crop matches the length of the growing season and the purpose for which the crop is to be grown. Variety selection trials to identify the best suitable varieties for given areas are frequently necessary.

When mean daily temperatures during the growing season are greater than 20°C, early grain varieties take 80 to 110 days and medium varieties 110 to 140 days to mature. When grown as a vegetable, these varieties are 15 to 20 days shorter. When mean daily temperatures are below 20°C, there is an extension in days to maturity of 10 to 20 days for each 0.5°C decrease depending on variety, and at 1.5°C the maize grain crop takes 200 to 300 days to mature. With mean daily temperature of 10 to 15°C maize is mostly grown as a forage crop because of the problem of seed set and grain maturity under cool conditions. For germination the lowest mean daily temperature is about 10°C, with 18 to 20°C being optimum. The crop is very sensitive to frost, particularly in the seedling stage but it tolerates hot and dry atmospheric conditions so long as sufficient water is available to the plant and temperatures are below 45°C. Temperature requirements, expressed as sum of mean daily temperatures, for medium varieties are 2500 to 3000 degree days, while early varieties require about 1800 and late varieties 3700 or more degree days.

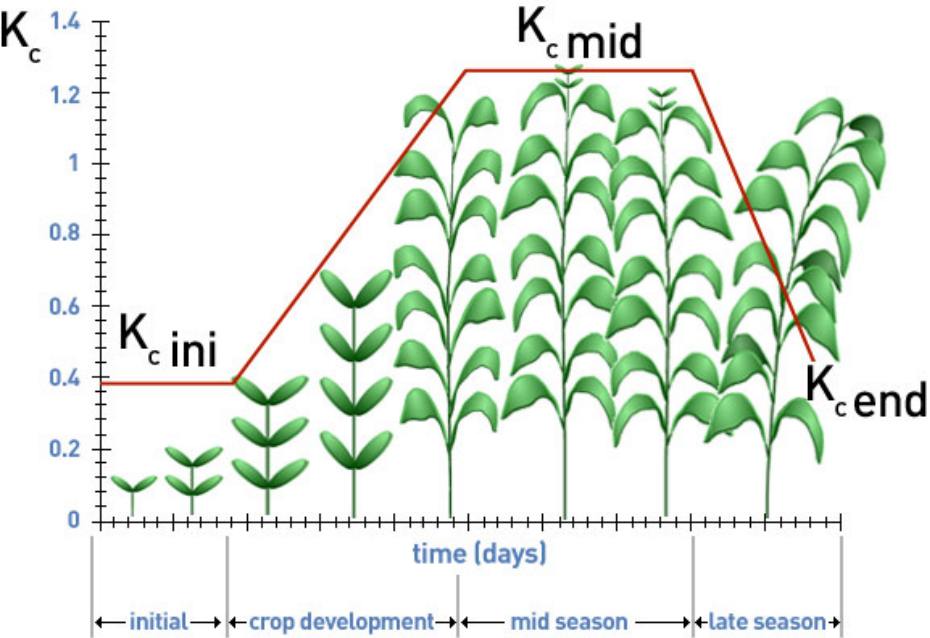
In respect of daylength, maize is considered to be either a day-neutral or a short-day plant. The growth of maize is very responsive to radiation. However, five or six leaves near and above the cob are the source of assimilation for grain filling and light must penetrate to these leaves. For optimum light interception, for grain production, the density index (number of plants per ha/row spacing) varies but on average it is about 150 for the large late varieties and about 500 for the small early varieties. Sowing methods and spacing vary, and fertility and water are decisive factors in choosing the optimum density in relation to light interception and highest yields. Plant population varies from 20000 to 30000 plants per ha for the large late varieties to 50000 to 80000 for small early varieties. Spacing between rows varies between 0.6 and 1 m. Sowing depth is 5 to 7 cm with one or more seeds per sowing point. When grown for forage, plant population is 50 percent higher.

The plant does well on most soils but less so on very heavy dense clay and very sandy soils. The soil should preferably be well-aerated and well-drained as the crop is susceptible to waterlogging. The fertility demands for grain maize are relatively high and amount, for high-producing varieties, up to about 200 kg/ha N, 50 to 80 kg/ha P and 60 to 100 kg/ha K. In general the crop can be grown continuously as long as soil fertility is maintained.

Maize is moderately sensitive to salinity. Yield decrease under increasing soil salinity is: 0% at ECe 1.7 mmhos/cm, 10% at 2.5, 25% at 3.8, 50% at 5.9 and 100% at ECe 10 mmhos/cm.

The graph below depicts the crop stages of maize, and the table summarises the main crop coefficients used for water management.

- CropWat
- CLIMWAT
- Diagnostic Tools for Investment
- FAO Nile
- ETo Calculator
- GAEZ
- GeoNetwork
- GLADIS
- HWSD
- Rapid Appraisal Procedure (RAP)
- SoiLEX
- SoiSTAT
- WaPOR
- WaterLex



Crop characteristic	Stages of Development					Plant date	Region
	Initial	Crop Development	Mid-season	Late	Total		
Stage length, days	30	50	60	40	180	April	East Africa (alt.)
	25	40	45	30	140	Dec/Jan	Arid Climate (alt.)
	20	35	40	30	125	June	Nigeria (humid)
	20	35	40	30	125	Oct./Dec	India (dry, cool)
	30	40	50	30	150	April	Spain (spr, sum)
	30	40	50	30	150	April	Calif.
	30	40	50	50	170	April	Idaho, USA
Depletion Coefficient, p	0.50	0.50	0.50	0.80	-		
Root Depth, m	0.30	>>	>>	1.00	-		
Crop Coefficient, Kc	0.30	>>	1.2	0.5	-		
Yield Response Factor, Ky	0.40	0.40	1.30	0.50	1.25		

Crop characteristic	Stages of Development				
	Initial	Crop Development	Mid-season	Late	Total
Stage length, days	25	40	40	35	135

	Stages of Development					
	Depletion Coefficient, p	0.50	0.50	0.50	0.80	-
	Root Depth, m	0.30	>>	>>	1.00	-
	Crop Coefficient, Kc	0.30	>>	1.2	0.5	-
	Yield Response Factor, Ky	0.40	1.0.40	1.30	0.50	1.25

Water Requirements

Water Supply And Crop Yield

Water Uptake

Irrigation Scheduling

Yield

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